m = 2 random-V curve effect at n = 19 with 32 subspaces (2026-10-25)

Run directory `research/koblitz-isogeny/reports/run-20261025T0530Z/`. Full note, section "m = 2, random V, n = 19 at 32 V": `research/notes/koblitz-isogeny/m3-presentation-vs-curve-32v-20261025.md`.

Question and design

Does the decomposition cost depend on the curve, or only on the factor-base presentation (the F2-subspace V), in the Gröbner regime with non-geometric V? The m = 3 arms covered geometric V with a linear-algebra solver only. Model: two-way crossed random effects, curves × subspaces; ICC_{curve} with a 2000-resample curve bootstrap (95 % CI). Pre-registered rule, unchanged: **H0** if every metric's upper bound is < 0.05; **H1** if any cost metric has ICC ≥ 0.10 with lower bound > 0; otherwise inconclusive. Metrics: `work_per_call` (reductions per call), `yield_per_Fm` (yield per $|F|^2$), `log_us_per_call`, `log_projected_solve`.

Change against the original n = 19 arm (128 curves × 4 random V): 32 random V per curve on the first 32 curves of the sample (seeded hash order, K_a included; operator statement). Class n = 19, a2 = 1, l = 9; m = 2 Gröbner; 400 full-group probes per cell; seeds 1–32. Host: shared container, no isolation level earned.

Data: `pres_19_1_l9_s1-32.jsonl`, 1408 rows = 512 original (128 curves × seeds 1–4) + 896 new (32 curves × seeds 5–32). Counted from the file: 1408 `bsgs_ok`, 0 inconsistent, none skipped; 124920 relations; 3.372 reductions per call (sum of `reductions` over sum of `groebner_calls`; new cells 3.373, original 3.371). The ICC keeps the 32 curves with every seed: 1024 cells. Every recorded `first_fall_hist` entry (4412) is at degree 3.

Result (ICC_{curve} [95 % CI], 32 curves × 32 V)

metric	ICC_{curve} [95 % CI]	subspace share
`work_per_call`	0.003 [0, 0.018]	0.002
`yield_per_Fm`	0.002 [0, 0.022]	0.000
`log_us_per_call`	0.000 [0, 0.009]	0.013
`log_projected_solve`	0.003 [0, 0.020]	0.002

Diagnostic: |F| 0.000 [0, 0.012]. The pins and roots diagnostics in the JSON are zero with zero variance (m = 2 has no pin step) and are not plotted.

Verdict by the pre-registered rule: **H0** (largest upper bound 0.0223). The m = 3 arms of `run-20261024T2200Z` had largest upper bounds 0.0205 (n = 31) and 0.0388 (n = 23).

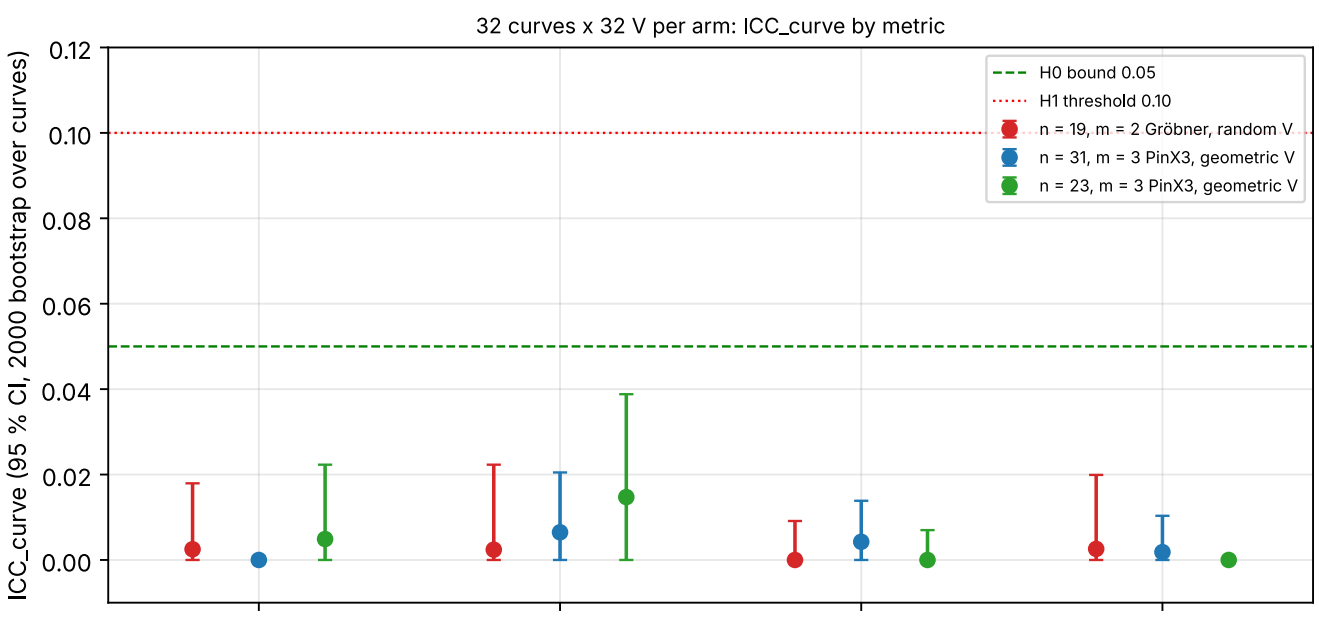


Figure 1: ICC_{curve} with 95 % CI by metric for the three 32 curves × 32 V arms: n = 19, m = 2 Gröbner, random V (this run); n = 31 and n = 23, m = 3 PinX3, geometric V (`run-20261024T2200Z`). "yield per |F|^m" is $|F|^2$ at m = 2 and $|F|^3$ at m = 3. Sources: `icc_n19_m2_rand_s1-32.json`, `icc_n31_m3_geom_s1-32.json`, `icc_n23_m3_geom_s1-32.json`. Dashed: 0.05; dotted: 0.10. Arms differ in solver, V type, class and metric definition, so the points are placed side by side, not pooled.

Detection limit (simulated verdict rates)

`power` rows at 32 V; 200 simulations per point, 32 curves, fitted additive normal model with a planted curve share. For this arm: P(H0) at true ICC 0 is 0.995–1.00 across the four metrics; P(H1) at 0.10 is 0.42 (work per call) and 0.42–0.49 across metrics; P(H1) at 0.20 is 0.98–0.995 (work per call 0.99). A curve share of 0.20 would almost surely have been called H1; 0.10 about half the time; a share between 0.05 and 0.10 is not excluded. Monte Carlo standard error near 0.5 is about 0.035.

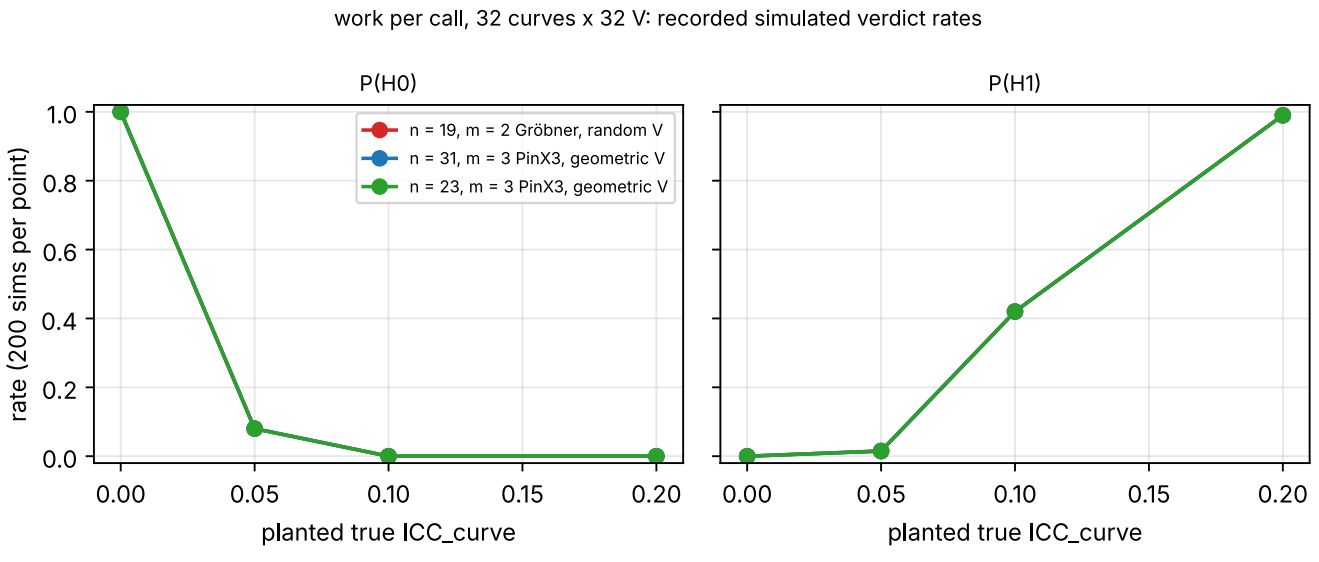


Figure 2: `work_per_call`: recorded P(H0) and P(H1) against the planted ICC at 32 V for the three arms (200 simulations per point). Same source files.

Limits

- One class (n = 19), m = 2 only, l = 9, 400 probes per cell, the 32 sampled curves. Other n, solvers and the first-fall quantity as a metric are not covered; the file has no first-fall variation (all entries at degree 3).
- H0 means no curve share detected above the stated bound. It is not a proof of curve independence.
- Timing is wall clock on a shared container without isolation. 128 of the 1024 analysed cells (seeds 1–4) are from the curve-order run whose timings an earlier note excluded as drift-confounded. Cell order of the new cells was not checked. Work per call and yield are deterministic.
- Additive model: a curve × V interaction is counted in the residual.
- Not run: n = 23, m = 2, random V at 32 V (about 900 s per cell, operator figure).
- Not a speed result; no cross-method comparison, leaderboard or `benchmarks/` entry; no validation-gate directory for these data.

Records

`experiments/koblitz_presentation/`: `pres_19_1_l9_s1-32.jsonl`, `icc_n19_m2_rand_s1-32.json`, `icc_n31_m3_geom_s1-32.json`, `icc_n23_m3_geom_s1-32.json`. Graph source: `plot.py`.